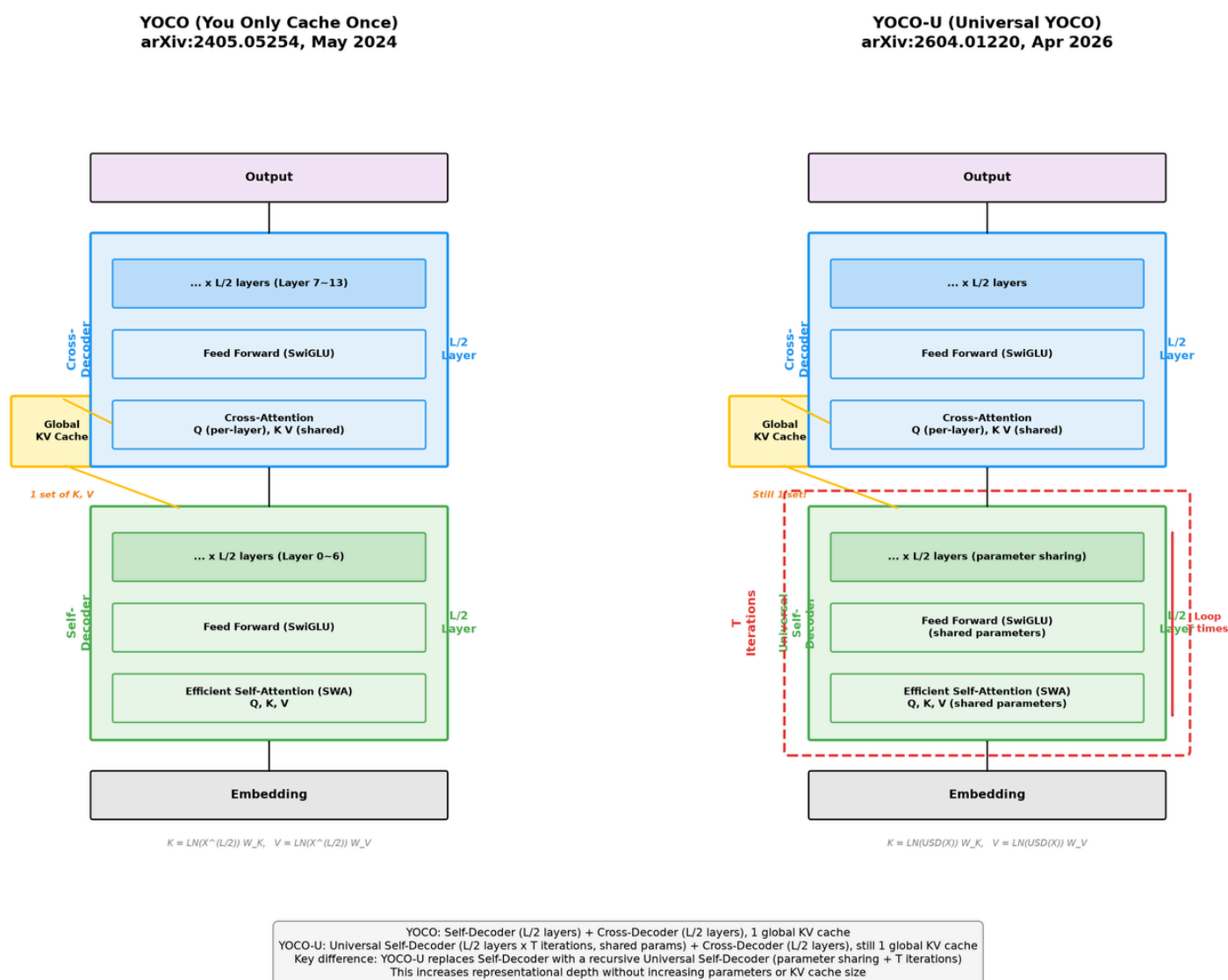


yoco-v3: yoco-moe-30b and yoco-u-moe-30b

1. 模型介绍 yoco-moe-30b和yoco-u-moe-30b 模型配置



2. 完成bitnet.cpp和llama.cpp对 yoco-moe-30b 和 yoco-u-moe-30b 模型的适配

代码块

```
1 YOCO_MOE_30B_A3B_V3 = ModelConfig(  
2     d_model=3072,  
3     d_ffn=9216,  
4     head=32,  
5     cross_head=32,
```

```
6     kv_head=8,
7     cross_kv_head=8,
8     head_dim=128,
9     n_layers=20,
10    qk_norm=False,
11    qk_rms_clip=True,
12    diff_v3=True,
13    weight_tying=True,
14    yoco_cross_layers=10,
15    yoco_window_size=512,
16    moe=True,
17    moe_ffn_dim=1280,
18    moe_expert_num=128,
19    moe_top_k=8,
20    d_shared_expert=1280,
21    moe_latent_dim=0,
22    moe_latent_norm=False,
23    bit_linear="attn_qkv,attn_out,attn_gate,yoco_kv,ffn",
24    moe_expert_bit_linear=True,
25    act_quant_method="adp_8bits",
26    weight_quant_method="TernarySEQ",
27    embproj=True,
28    emb_proj_norm=True,
29    int8_embedding=True,
30    int8_lm_head=True,
31 )
```

3. 模型精度测试

3.1 YOCO-MoE模型精度测试

```

root@amax:/home/huangxin# python /home/huangxin/code_list/bitnet-moe-script/verify_precision_script/verify_precision_yoco_moe_30b_a3b_v3.py
=====
Precision Verification: TernarySEQ (PyTorch) vs I2_S (BitNet TL2)
Model: YOCO-MoE-30B-A3B-V3
=====

Both paths use per-row quantization:
alpha = max(|W_row|)
ternary = round(W/alpha * 1.5) / 1.5 -> {-1, 0, +1}
dequant = ternary * alpha

=====
Layer      Shape      MaxDiff      MeanDiff      CosSim      RelErr
-----
Q proj     (8192, 3072) 0.000000    0.000000000    1.000000    0.000000000
O proj     (3072, 4096) 0.000000    0.000000000    1.000000    0.000000000
MoE gate_up (1280, 3072) 0.000000    0.000000000    1.000000    0.000000000
MoE down   (3072, 1280) 0.000000    0.000000000    1.000000    0.000000000
K/V proj   (1024, 3072) 0.000000    0.000000000    1.000000    0.000000000
Self-attn  (3072, 3072) 0.000000    0.000000000    1.000000    0.000000000
Router     (128, 3072)  0.000000    0.000000000    1.000000    0.000000000
=====

Average cosine similarity: 1.000000
Average max diff:         0.000000

✅ PASS: TernarySEQ and I2_S paths are numerically identical
root@amax:/home/huangxin#

```

3.2 YOCO-U-MoE模型精度测试

```

root@amax:/home/huangxin# python /home/huangxin/code_list/bitnet-moe-script/verify_precision_script/verify_precision_yoco_u_moe_30b_a3b_v3.py
=====
Precision Verification: TernarySEQ (PyTorch) vs I2_S (BitNet TL2)
Model: YOCO-U-MoE-30B-A3B-V3 (T=3 universal loop)
=====

Both paths use per-row quantization:
alpha = max(|W_row|)
ternary = round(W/alpha * 1.5) / 1.5 -> {-1, 0, +1}
dequant = ternary * alpha

Note: YOCO-U shares self-decoder weights across T=3 iterations.
Tensor shapes are identical to YOCO-MoE-30B-A3B-V3.

=====
Layer      Shape      MaxDiff      MeanDiff      CosSim      RelErr
-----
Q proj     (8192, 3072) 0.000000    0.000000000    1.000000    0.000000000
O proj     (3072, 4096) 0.000000    0.000000000    1.000000    0.000000000
MoE gate_up (1280, 3072) 0.000000    0.000000000    1.000000    0.000000000
MoE down   (3072, 1280) 0.000000    0.000000000    1.000000    0.000000000
K/V proj   (1024, 3072) 0.000000    0.000000000    1.000000    0.000000000
Self-attn  (3072, 3072) 0.000000    0.000000000    1.000000    0.000000000
Router     (128, 3072)  0.000000    0.000000000    1.000000    0.000000000
=====

Average cosine similarity: 1.000000
Average max diff:         0.000000

✅ PASS: TernarySEQ and I2_S paths are numerically identical
root@amax:/home/huangxin#

```

4. 模型精度测试

4.1 服务器硬件信息

代码块

- 1 CPU: Intel Xeon Platinum 8358P (Ice Lake-SP)
- 2
- 3 内存带宽:
- 4 - Xeon 8358P 支持 DDR4-3200 (8 通道)
- 5 - 单 CPU 理论峰值带宽: $8 \times 3200 \text{ MHz} \times 8 \text{ bytes} = 204.8 \text{ GB/s}$
- 6 - 你有 2 路 CPU, 系统总理论峰值带宽约 409.6 GB/s

4.1 YOCO-MoE模型性能测试

- 不开启TL2
 - F16.GGUF

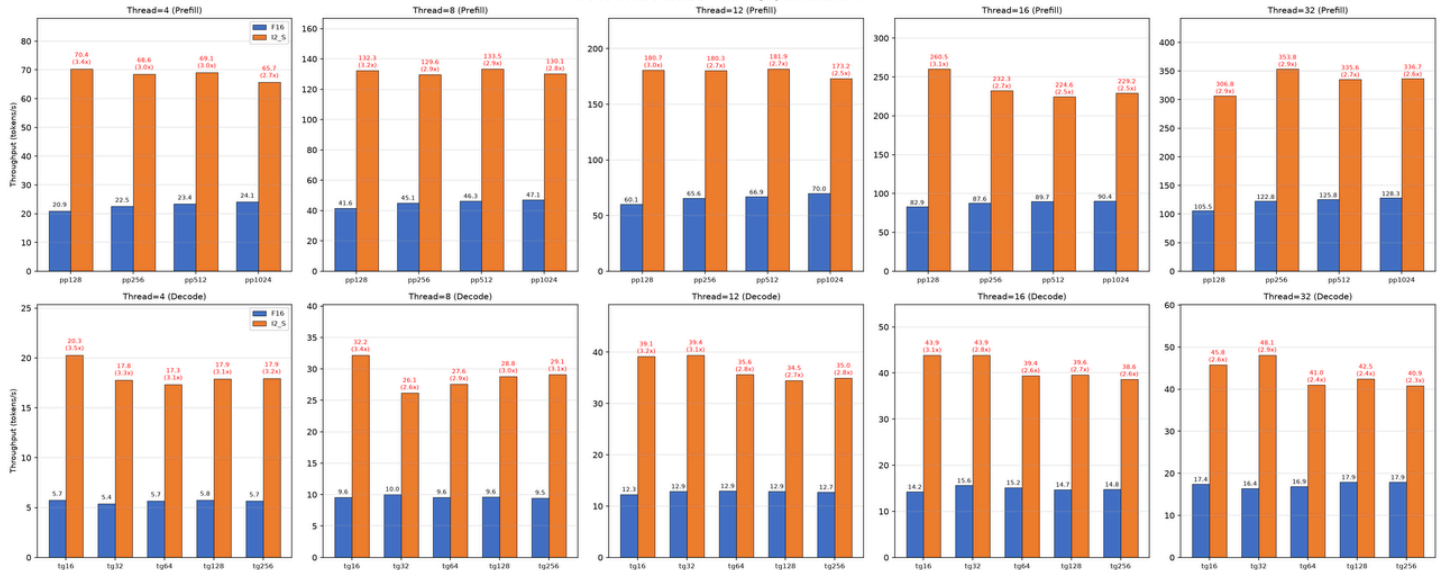
Model Thread Number	Thread = 1	Thread = 2	Thread = 4	Thread = 8	Thread = 12	Thread = 16	Thread = 32
YOCO-MoE-30B.A3B	(token s / s)	(token s / s)	(token s / s)	(token s / s)	(token s / s)	(tokens / s)	(tokens / s)
pp128	-	-	20.93 ± 0.01	41.63 ± 2.60	60.11 ± 7.31	82.89 ± 0.66	105.55 ± 26.61
pp256	-	-	22.53 ± 0.55	45.06 ± 3.37	65.63 ± 4.41	87.57 ± 8.95	122.81 ± 12.45
pp512	-	-	23.41 ± 0.49	46.26 ± 1.42	66.89 ± 1.87	89.70 ± 3.97	125.85 ± 5.68
pp1024	-	-	24.09 ± 0.20	47.06 ± 1.03	69.95 ± 0.19	90.42 ± 2.32	128.33 ± 3.94
tg16	-	-	5.73 ± 0.80	9.59 ± 1.81	12.30 ± 2.13	14.23 ± 3.47	17.42 ± 0.03
pp32	-	-	5.38 ± 0.06	10.02 ± 1.01	12.86 ± 0.81	15.61 ± 0.72	16.43 ± 1.48
tg64	-	-	5.66 ± 0.17	9.58 ± 0.08	12.93 ± 0.76	15.17 ± 0.96	16.90 ± 1.33
tg128	-	-	5.75 ± 0.03	9.61 ± 0.22	12.87 ± 0.30	14.73 ± 0.54	17.93 ± 0.21
tg256	-	-	5.65 ± 0.06	9.47 ± 0.06	12.70 ± 0.06	14.77 ± 0.20	17.91 ± 0.13

- I2_S.GGUF

Model Thread Number	Thread = 1	Thread = 2	Thread = 4	Thread = 8	Thread = 12	Thread = 16	Thread = 32
--------------------------	---------------	---------------	---------------	---------------	----------------	----------------	----------------

YOCO-MoE-30B.A3B	(token s / s)	(token s / s)	(token s / s)	(token s / s)	(token s / s)	(tokens / s)	(tokens / s)
pp128	-	-	70.39 ± 6.27	132.31 ± 14.19	180.65 ± 25.66	260.51 ± 2.21	306.75 ± 82.33
pp256	-	-	68.58 ± 6.27	129.64 ± 19.55	180.26 ± 30.89	232.34 ± 47.14	353.81 ± 5.89
pp512	-	-	69.13 ± 0.96	133.48 ± 13.74	181.88 ± 11.75	224.57 ± 28.15	335.62 ± 56.93
pp1024	-	-	65.72 ± 0.85	130.12 ± 1.23	173.25 ± 3.21	229.23 ± 10.72	336.66 ± 30.70
tg16	-	-	20.29 ± 0.21	32.21 ± 0.11	39.15 ± 0.41	43.85 ± 0.55	45.80 ± 0.40
pp32	-	-	17.79 ± 4.07	26.15 ± 5.49	39.44 ± 0.17	43.90 ± 0.11	48.12 ± 0.32
tg64	-	-	17.33 ± 2.26	27.56 ± 4.29	35.62 ± 7.30	39.39 ± 8.87	40.99 ± 8.69
tg128	-	-	17.91 ± 0.09	28.79 ± 3.00	34.47 ± 4.61	39.60 ± 4.77	42.47 ± 3.28
tg256	-	-	17.94 ± 0.12	29.09 ± 0.04	34.95 ± 0.10	38.63 ± 0.61	40.85 ± 0.22

YOCO-MoE-30B.A3B - Throughput (tokens/s)



4.1 YOCO-U-MoE模型性能测试

- 不开启TL2

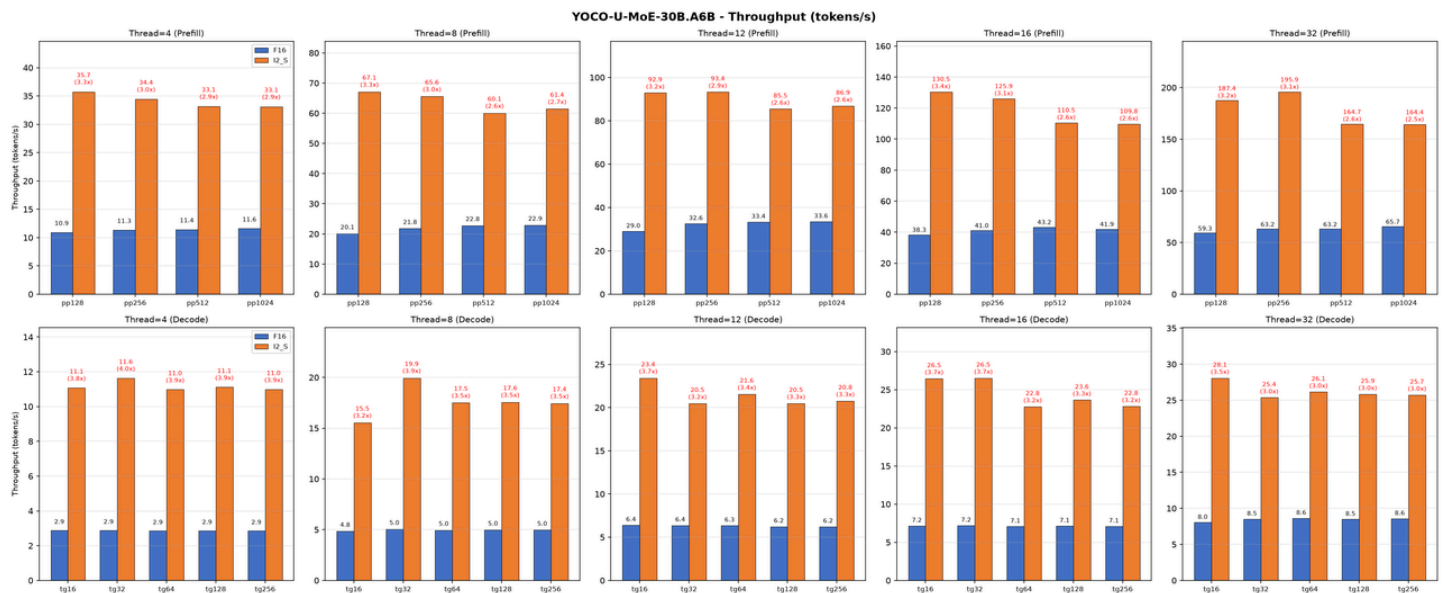
◦ F16.GGUF

Model Thread Number YOCO-U-MoE- 30B.A6B	Thread = 1 (token s / s)	Thread = 2 (token s / s)	Thread = 4 (token s / s)	Thread = 8 (token s / s)	Thread = 12 (token s / s)	Thread = 16 (tokens / s)	Thread = 32 (tokens / s)
pp128	-	-	10.92 ± 0.03	20.06 ± 0.69	29.02 ± 1.93	38.27 ± 3.87	59.27 ± 4.96
pp256	-	-	11.32 ± 0.12	21.76 ± 0.33	32.59 ± 0.10	40.98 ± 0.65	63.20 ± 2.67
pp512	-	-	11.43 ± 0.17	22.79 ± 0.33	33.38 ± 0.18	43.17 ± 0.63	63.16 ± 0.25
pp1024	-	-	11.59 ± 0.10	22.92 ± 0.14	33.56 ± 0.08	41.86 ± 0.39	65.67 ± 0.81
tg16	-	-	2.88 ± 0.18	4.84 ± 0.53	6.38 ± 0.69	7.16 ± 1.07	8.04 ± 0.71
pp32	-	-	2.88 ± 0.07	5.05 ± 0.19	6.36 ± 0.27	7.20 ± 0.61	8.52 ± 0.65
tg64	-	-	2.85 ± 0.05	4.95 ± 0.05	6.33 ± 0.20	7.08 ± 0.21	8.64 ± 0.11
tg128	-	-	2.86 ± 0.04	4.97 ± 0.11	6.23 ± 0.03	7.13 ± 0.11.	8.49 ± 0.12
tg256	-	-	2.85 ± 0.01	4.96 ± 0.05	6.21 ± 0.03	7.10 ± 0.01	8.58 ± 0.05

◦ I2_S.GGUF

Model Thread Number YOCO-U-MoE- 30B.A6B	Thread = 1 (token s / s)	Thread = 2 (token s / s)	Thread = 4 (token s / s)	Thread = 8 (token s / s)	Thread = 12 (toke ns / s)	Thread = 16 (tokens / s)	Thread = 32 (tokens / s)
pp128	-	-	35.74 ± 3.52	67.11 ± 9.93	92.92 ± 15.34	130.48 ± 2.99	187.45 ± 4.62

pp256	-	-	34.42 ± 0.65	65.58 ± 4.10	93.37 ± 7.94	125.94 ± 12.19	195.92 ± 8.56
pp512	-	-	33.15 ± 1.05	60.07 ± 2.07	85.51 ± 4.18	110.48 ± 5.25	164.68 ± 9.65
pp1024	-	-	33.10 ± 0.12	61.43 ± 0.76	86.91 ± 2.31	109.82 ± 3.07	164.44 ± 3.01
tg16	-	-	11.08 ± 2.79	15.54 ± 4.01	23.42 ± 0.08	26.46 ± 0.09	28.08 ± 0.10
pp32	-	-	11.64 ± 1.63	19.94 ± 0.26	20.47 ± 4.95	26.53 ± 0.28	25.37 ± 5.08
tg64	-	-	10.98 ± 0.10	17.50 ± 2.14	21.55 ± 1.58	22.78 ± 3.01	26.15 ± 2.68
tg128	-	-	11.12 ± 0.38	17.56 ± 0.08	20.47 ± 0.09	23.65 ± 1.65	25.85 ± 1.45
tg256	-	-	10.99 ± 0.09	17.44 ± 0.04	20.76 ± 0.87	22.81 ± 0.84	25.74 ± 0.74



Todo list

1. 在bitnet跑yoco-moe-dense-3b, yoco-moe-dense-6b
2. 在llama.cpp跑qwen3.8-27b-dense, 同时弄成bitnet看吞吐和性能
3. 在llama.cpp跑qwen-moe-27b-a3b去看decode吞吐, 对比1, 去证明我的达到了sota

4. 看推理时总内存消耗